Formal Modeling and Analysis of Metric-Timed Systems and Multi-Agent Normative Specifications of Actions

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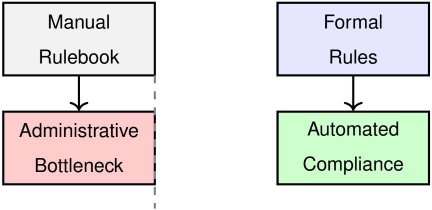
Outline

Introduction and Context

The Cost of Normative Compliance

Motivation:

- Normative systems govern behavior via **ought-to-do rules**.
- Manual verification creates a heavy economic burden.
- Real-world rules involve strict **time constraints** and **distributed agents**.



The Problem:

- Fragments of laws often **conflict** (ambiguity).
- Accountability fails when

Research Directions

RD1 / Chapter 2: Static Conflict Analysis

How can we formally detect, measure, and explain conflicts between time-constrained obligations and prohibitions?

RD2 / Chapter 3: Blame Attribution in multi-agent setting

How do we formalize and quantify responsibility when collaborative execution of norms fails in a multi-agent setting?

Methodology: Micro Metric-Time Normative Logic (μMTNL)

Syntax: restricted to essential structures for practitioners

Focus on precisely timed **Obligations** (O) and **Prohibitions** (F).

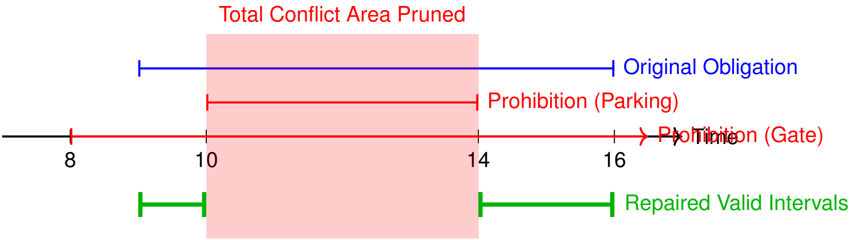
$$\phi ::= O^{\mathfrak{I}}(\mathrm{a}) \mid F^{\mathfrak{I}}(\mathrm{a}) \mid \phi \sqcap \phi \mid \phi \sqcup \phi$$

Where $\mathfrak{I}$ is a canonical set of disjoint time intervals.

The Analysis Pipeline: From sat to conflict characterisation and repair

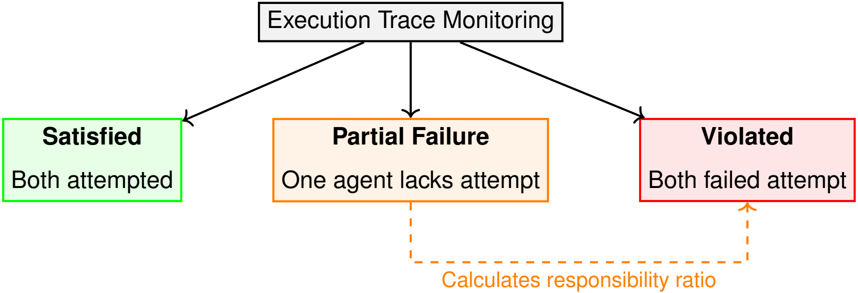
1. Sat resolution: Determine if a specification is satisfiable.
2. Conflict static definition: Deontic and Ontic conflicts.
3. Conflict measurement: Quantify the degree of conflict using a novel metric.
4. Conflict characterization: If not fully conflicting, eliminate the minimal unsatisfiable subsets (MUSes) to explain the conflict

Discussion: Visualizing Conflict Resolution



Key Results: The Blame Attribution Procedure

- Formalized **Obligation task of an agent as**: *"Agent x has to accomplish task a ..."*
- Quantified blame based on the lack of *attempt* over a defined duration.



Conclusion

Core Takeaway

Restricting logical expressivity in a principled manner enables stronger, operational forms of automated reasoning about normative systems.

Summary of Deliverables:

1. MTNL: Formal pipeline for timed conflict detection.
2. **Automated Repair**: Algorithms balancing conciseness and conflict elimination.
3. cDL: Deepened dynamic analysis via blame attribution in multi-agent task collaboration and in a perpetual manner.

Future Work

Toward a Unified Formalism (MTTACNL)

Mixing the timed μMTNL with the multi-agent aspect of TACNL and getting both conflict and monitoring algoirthms for this setting.

- **ITP integration** Integrating the logics into interactive theorem provers (e.g., Roq, Lean).
- **Moving beyond Discrete actions:** Actions with directions and continous states(ought-to-be).